

How Novel Is a “Held-Out” Reaction? A Within-Training Control for Similarity Audits of the ORDerly Condition-Prediction Benchmark

David Lindgreen

October 9, 2026

Version 0.1.0 Preprint, not peer reviewed

Abstract

Reaction-condition benchmarks are usually split so that no test reaction shares its exact inputs with training, which is taken to mean the test set is novel. ORDerly [1] follows this practice. We audited its “reaction string, delete rare” split (625,697 training and 65,445 test reactions) and found that 5.78% of test products, and 80.84% of test product scaffolds, also occur in training, and that 60.5% of a 1,000-row test sample has a training product at Morgan–Tanimoto similarity ≥ 0.70 . Such numbers are uninterpretable without a reference: how much overlap would any hold-out of this data show? We supply one by applying the identical measurements to the training set itself, holding out one reaction at a time against rows with a different input key. The control gives 8.72% identity overlap, 81.63% scaffold overlap and 60.2% at ≥ 0.70 . The test split’s similarity to training is therefore what a random hold-out of this redundant corpus produces: the redundancy belongs to the dataset, not to the released split. The frozen audit and four frequency baselines reproduce exactly. Neural-model results could not be reproduced because the final checkpoints are not public.

1 Question

A benchmark’s difficulty depends on how different test examples are from training examples. Exact-key separation is necessary but weak: chemically near-identical products can sit on both sides. The repository asks whether ORDerly’s large reported accuracy change survives an exact reproduction and which data decisions make the benchmark easier. The earlier audit measured test-to-training overlap but gave no baseline for it. This paper adds that baseline.

2 Frozen reproduction

Inputs are the pinned ORDerly benchmark and supplement (checksums committed; the 1.46 GB archive matched its recorded MD5). The four top-3 frequency baselines were recomputed from the official reference function and agree with the published figures: 20.24% and 19.55% for the reaction-string variants (published about 20%) and 52.22% and 51.57% for the trusted-label variants (published about 52%). Re-running the prespecified similarity audit reproduced the committed result with no differences.

Test-to-training measurement (frozen audit)	Value
Product identity occurs in training	5.78% (3,784 of 65,444)
Product scaffold occurs in training	80.84% (51,617 of 63,852)
Sampled max Tanimoto, median / mean	0.734 / 0.732
Sample at Tanimoto ≥ 0.70	60.5% (57.4–63.5)
Sample at Tanimoto ≥ 0.80	30.0%
Sample at Tanimoto ≥ 0.90	11.7%
Sample at Tanimoto = 1.00	8.4%

Table 1: Frozen similarity audit; intervals are Wilson 95%.

3 Post-hoc: a within-training control

For every training reaction we asked the same questions as for a test reaction, using as references only the other training rows whose exact input key (two reactants and the product string) differs from its own. This mirrors the split’s defining property that no test key appears in training. Canonicalisation, Bemis–Murcko scaffolds and 2,048-bit radius-2 Morgan fingerprints are identical to the frozen audit. The similarity measurement uses a seeded random sample of 1,000 training rows. This analysis was not preregistered and all of it is reported. At the scaffold level and across

Measurement	Test vs training	Training vs other training
Product identity overlap	5.78%	8.72% (54,530 of 625,695)
Scaffold overlap	80.84%	81.63%
Max Tanimoto, median	0.734	0.729
Max Tanimoto, mean	0.732	0.738
≥ 0.70	60.5%	60.2% (57.1–63.2)
≥ 0.80	30.0%	31.8%
≥ 0.90	11.7%	14.0% (12.0–16.3)
= 1.00	8.4%	11.5% (9.7–13.6)

Table 2: Same measurements on the released test split and on the training set held out row by row.

the similarity distribution up to 0.80 the two columns are indistinguishable. The control is *higher* than the test split for exact product identity (8.72% against 5.78%) and for near-identical similarity (≥ 0.90 and = 1.00), so the released split is, if anything, somewhat less redundant with training than an in-sample hold-out. The mechanism is not determined here; the sampling intervals for the similarity rows are wide enough that only the identity-level difference is clear.

4 Discussion

A high overlap figure does not by itself indict a split. For this corpus, which contains many near-analogue reactions from patents, an honest random hold-out already has most test scaffolds and about 60% of products with a close neighbour in training. The finding is therefore about the *dataset*: accuracy on any split of it partly measures recall of nearby chemistry. That is a statement about what the benchmark can establish, not a claim that ORDERly leaked information. A split that is substantially more novel than this control (by scaffold or similarity-based clustering) would be a different and harder benchmark, and the control gives a concrete target for what “more novel” must exceed.

5 Limitations

Fingerprint similarity of 1.0 is not molecular identity. The similarity comparison rests on 1,000-row samples. Exact-key separation in the control follows the audit’s key definition, not patent family, which is unavailable. Only the “reaction string, delete rare” variant was audited for similarity. The neural-model accuracies could not be reproduced (checkpoints and prediction files are not public), so nothing here bears on them; no claim is made about prospective performance or wet-lab outcomes.

Reproducibility and AI assistance

Code, frozen audits, the control and tests:

<https://github.com/lindgreendavid/reaction-integrity-lab>

Interactive laboratory:

<https://lindgreendavid.github.io/reaction-integrity-lab/>

The analysis code, tests and drafts of this text were produced with substantial assistance from an AI system (Claude, Anthropic); the author is responsible for the content.

References

- [1] D. S. Wigh, J. Arrowsmith, A. Pomberger, K. C. Felton, A. A. Lapkin, ORDERly: Data sets and benchmarks for chemical reaction data, *J. Chem. Inf. Model.* **64**(9), 3790–3798 (2024), doi:10.1021/acs.jcim.4c00292.